

EMERSON EIMS BUILDING SUITE PRO

PDF KE

Location: Nairobi, Kenya
Total Area: 220 m² | Units: 1 | Storeys: 2
Architectural Style: Modern
Date: 27 April 2026

PROJECT INFORMATION	
Project ID	59c2fac2-2d56-4b0a-9617-b178ef5eb0bd
Location	Nairobi, Kenya
Total Area	220 m²
Units / Apartments	1
Storeys	2
Bedrooms per Unit	4
Foundation Type	Strip + Ground Beam
Style	Modern
Report Generated	2026-04-27 03:01

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VALIDATION ENGINE — AUDIT TRAIL

■ 1 BLOCKER(S) — DO NOT SUBMIT

• Kenya statutory IDs missing: Plot / cadastral reference; Architect (must be registered with BORAQS); Structural Engineer (must be registered with EBK); Quantity Surveyor (must be registered with BORAQS). Required for permit submission to County Government Building Plans Section.

1 auto-correction(s) applied with code citation:

Field	Original	Auto-corrected to	Reason / Code citation
Foundation	(unspecified)	strip + ground beam	2-storey → strip + ground beam (BS 8004 §7)

Compliance Status: **BLOCKED — DO NOT SUBMIT**

1. DESIGN BRIEF & DESIGN CRITERIA

Item	Value
Project name	PDF KE
Location	Nairobi, Kenya
Total floor area	220.0 m²
No. of storeys / floor count	2 (single source of truth)
Dwelling units	1
Bedrooms / unit	4
Architectural style	Modern
Foundation	STRIP + GROUND BEAM
Currency	KES
Plot / LR No.	— REQUIRED —
Coordinates (lat,lng)	-1.2921,36.8219
Title Deed	—
Architect (BORAQS)	— REQUIRED —
Structural Engineer (EBK)	— REQUIRED —
Quantity Surveyor (BORAQS)	— REQUIRED —
Client	—

Design loads (BS EN 1991-1-1 Cat A — residential):

Action	Value (kN/m²)	Reference
Permanent action g_k (slab + finishes + partitions)	7.00	EC1 Table 6.1
Variable action q_k	2.00	EC1 Table 6.2 Cat A
Design ULS $w = 1.35 g_k + 1.5 q_k$	12.45	EC0 §6.4.3.2
Design SLS $w = g_k + q_k$	9.00	EC0 §6.5.3

2. STRUCTURAL DESIGN CALCULATIONS

Calculations to BS EN 1992-1-1 (EC2) using $f_{ck} = 30 \text{ MPa}$, $f_{yk} = 500 \text{ MPa}$, cover per exposure class XC1 internal / XC3 external. All members hand-verified against simplified Eurocode 2 design rules; final detailing to be reviewed by the Structural Engineer of Record (EBK-registered) before issue for construction.

2.1 Slab design (one/two-way RC slab)

Parameter	Value
Short span L_x	9.57 m
Long span L_y	11.49 m
Behaviour	Two-way
Slab thickness h	370 mm (span/depth = 26)
Effective depth d	339.0 mm
Cover	25 mm (XC1)
Design moment M	70.76 kN·m / m
As required	505.0 mm ² / m
As minimum (EC2 §9.3.1.1)	481.0 mm ² / m
As provided	505.0 mm ² / m
Reinforcement	T12 @ 200 c/c BW

2.2 Beam design (simply-supported RC beam)

Parameter	Value
Clear span L	5.74 m
Beam section $b \times h$	237 × 475 mm
Effective depth d	429.0 mm
Cover	30 mm
Service load w	62.41 kN/m
Design moment M	257.5 kN·m
Design shear V	179.3 kN
As required (tension)	1452.0 mm ²
Main steel	8 no. T16 bot · 4 no. T16 top
Shear links	T8 @ 300 c/c

2.3 Column design (axially loaded RC column)

Parameter	Value
Axial load N (ULS)	619.0 kN
Section	225 × 225 mm
Concrete grade	C25/30 ($f_{ck} = 30 \text{ MPa}$)
Steel grade	$f_{yk} = 500 \text{ MPa}$
Steel ratio ρ	0.020 (2.0 %)
As required	1012.0 mm ²
Main steel	6 no. T16
Links	T8 @ 192 c/c

2.4 Foundation design (per BS 8004 / EN 1997-1)

Parameter	Value
Foundation type	STRIP + GROUND BEAM
Plan size	B = 1040 mm wide × Df = 1200 mm deep
Thickness	300 mm
Concrete grade	C25/30
Cover	75 mm (XC2 ground contact)
Safe bearing pressure (presumed)	150 kPa
Column load to foundation	619.0 kN
Reinforcement	T12 @ 150 c/c BW bottom + T10 @ 200 distribution top

■ The bearing pressure shown is a BS 8004 Table 1 *presumed* value pending a site investigation. A signed Geotechnical Investigation Report (boreholes, SPT, allowable bearing capacity) must be obtained from a geotechnical engineer before construction. EIMS records this gap honestly rather than fabricating SPT values.

3. BAR BENDING SCHEDULE (BS 8666)

Schedule of all reinforcement bars with mark, diameter, shape code, length per bar, number off, total length and weight. Steel grade 500B. Bend radii and bar shapes comply with BS 8666:2020 Table 2. Quantities to be verified by the Engineer before cutting and bending.

Bar Mark	Dia.	Shape Code	Length per bar (m)	No. off	Total length (m)	Weight (kg)
S1	T12	00	15.13	148	2239.59	1988.8
B1-bot	T16	00	6.24	108	674.41	1064.9
B1-top	T16	37	6.24	36	224.80	355.0
B1-link	T8	51	0.87	342	298.91	118.1
C1	T16	00	3.50	54	189.00	298.4
C1-link	T8	51	0.60	162	97.20	38.4
F1	T12	00	3.00	108	324.00	287.7
					TOTAL	4151.3

Total reinforcement steel: 4,151 kg $\approx$ 4.15 tonnes

4. MEP DESIGN — ELECTRICAL, PLUMBING & DRAINAGE

Electrical design per IEC 60364 / KS IEC 60364. Plumbing & sanitary drainage per BS EN 12056-2 and Kenya NCC Building Code SEHS. Septic tank to NEMA EIA Regulations 2003.

4.1 Electrical load schedule

Load description	Value
Lighting (LED) @ 5 W/m ²	1100 W
Small power sockets @ 25 W/m ²	5500 W
Cooker / hob (per unit)	6000 W
Water heaters	6000 W
Air conditioning (split units)	6000 W
TOTAL CONNECTED LOAD	24,600 W
Diversity factor	0.65
DESIGN MAXIMUM DEMAND	15,990 W (16.8 kVA)
Main breaker (single phase 230 V)	75 A

4.2 Final-circuit schedule

Circuit	Load	Cable	Protection	RCD
L1-Ln (Lighting)	1100 W	1.5 mm ² T+E	10 A B-curve MCB	—
P1-Pn (Sockets)	5500 W	2.5 mm ² T+E	20 A B-curve MCB	30 mA RCD
C (Cooker)	6000 W	6 mm ² T+E	32 A B-curve MCB	—
W (2× water heater)	6000 W	4 mm ² T+E	25 A B-curve MCB	30 mA RCD
A1-A4 (AC splits)	6000 W	2.5 mm ² T+E	16 A C-curve MCB	—

4.3 Single-line diagram (text representation)

KPLC supply ■ kWh meter ■ Main isolator 75 A ■ Main DB (Type-B SPD)
■ Lighting circuits L1...Ln (1.5 mm² · 10 A)
■ Socket circuits P1...Pn (2.5 mm² · 20 A · 30 mA RCD)
■ Cooker C (6 mm² · 32 A)
■ Water heaters W (4 mm² · 25 A · 30 mA RCD)
■ AC splits A1...An (2.5 mm² · 16 A)
■ Earth busbar ■ copper earth-rod ≤ 1 Ω

4.4 Plumbing fixture schedule

Fixture	Qty
Water Closet (WC)	2
Wash-Hand Basin	2
Shower / Bath	2
Kitchen Sink	1
Laundry Sink	1
External Bib Tap	2

4.5 Drainage gradient table (BS EN 12056-2)

Pipe	Slope range	Min fall
100 mm soil pipe (WC)	1:40 to 1:80	≥ 1.00 m fall per 1 m
75 mm waste	1:40	0.025 m / m

50 mm waste	1:40	0.025 m / m
32 mm WHB / sink	1:20	0.050 m / m

4.6 Septic tank & soak-pit sizing

Parameter	Value
Design occupancy	8 persons (BS 8233 — bedrooms × 2)
Daily wastewater flow (180 L/p/d)	1,440 L/day
Septic tank volume (3-day retention)	4.32 m ³
Suggested tank dimensions	2.5 × 1.2 × 1.44 m (L×W×D)
Soak-pit area (porous soil, 30 L/m ² /d)	48.0 m ²
Soak-pit diameter (circular)	7.82 m

■ Final septic-system geometry must be confirmed by an on-site percolation test. A NEMA Environmental Impact Assessment (EIA) project report is required for systems > 50 persons or sites near surface watercourses.

5. BILL OF QUANTITIES (KQS-2025)

Contractor-grade BOQ in KES, structured by trade per the Kenya Quantity Surveyors Standard Method of Measurement. Rates are KQS-2025 indicative urban rates; tenderers shall price all items. All quantities to be verified on site.

A — PRELIMINARIES

Description	Unit	Qty	Rate	Amount
Site mobilisation, hoarding, signage	item	1.00	KSh 180,000	KSh 180,000
Insurance & NCA Levy	item	1.00	KSh 220,000	KSh 220,000
Site office + storage	item	1.00	KSh 95,000	KSh 95,000
			Trade subtotal	KSh 495,000

B — SUBSTRUCTURE

Description	Unit	Qty	Rate	Amount
Site clearance & strip topsoil	m²	286.00	KSh 120	KSh 34,320
Bulk excavation foundation trenches	m³	66.00	KSh 650	KSh 42,900
Hardcore filling 200 mm	m²	220.00	KSh 850	KSh 187,000
Lean concrete blinding 50 mm	m³	11.00	KSh 14,000	KSh 154,000
RC foundation C25/30	m³	66.00	KSh 26,000	KSh 1,716,000
			Trade subtotal	KSh 2,134,220

C — SUPERSTRUCTURE

Description	Unit	Qty	Rate	Amount
RC columns C25/30	m³	3.19	KSh 32,500	KSh 103,655
RC beams C25/30	m³	23.28	KSh 30,500	KSh 710,070
RC suspended slabs C25/30	m³	162.80	KSh 28,500	KSh 4,639,800
Reinforcement bars (BBS)	kg	4,151.30	KSh 165	KSh 684,964
Formwork to soffits & sides	m²	1,135.62	KSh 1,450	KSh 1,646,652
200 mm machine-cut block walling	m²	355.98	KSh 1,800	KSh 640,760
Plaster both sides	m²	711.96	KSh 1,100	KSh 783,151
			Trade subtotal	KSh 9,209,052

D — ROOFING

Description	Unit	Qty	Rate	Amount
Timber roof trusses & purlins	m²	220.00	KSh 2,200	KSh 484,000
0.6 mm box-profile coloured iron sheet	m²	231.00	KSh 1,450	KSh 334,950
9 mm gypsum ceiling on softwood brandering	m²	440.00	KSh 1,850	KSh 814,000
			Trade subtotal	KSh 1,632,950

E — FINISHES

Description	Unit	Qty	Rate	Amount
Floor ceramic / porcelain tiling	m²	374.00	KSh 2,400	KSh 897,600
Wall tiling (wet areas)	m²	79.20	KSh 2,200	KSh 174,240
Internal paint (Crown / Plascon emulsion)	m²	569.56	KSh 520	KSh 296,173
External paint (weather-shield)	m²	284.78	KSh 720	KSh 205,043

			Trade subtotal KSh 1,573,056
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F — JOINERY

Description	Unit	Qty	Rate	Amount
Internal flush doors with hardware	each	7.00	KSh 18,000	KSh 126,000
External main doors hardwood	each	2.00	KSh 42,000	KSh 84,000
Aluminium-framed sliding windows	m²	33.00	KSh 9,500	KSh 313,500
			Trade subtotal	KSh 523,500

G — ELECTRICAL

Description	Unit	Qty	Rate	Amount
Mains supply, DB, MCB, RCDs (per IEC 60364)	item	1.00	KSh 165,000	KSh 165,000
Lighting points & circuits (~1100 W)	point	39.60	KSh 2,200	KSh 87,120
Socket outlets (~5500 W)	point	26.40	KSh 2,400	KSh 63,360
			Trade subtotal	KSh 315,480

H — PLUMBING

Description	Unit	Qty	Rate	Amount
Cold + hot water reticulation, all fixtures	item	1.00	KSh 285,000	KSh 285,000
Septic tank + soak-pit (4.32 m³ + Ø7.82 m)	item	1.00	KSh 195,000	KSh 195,000
Stormwater drainage & rain-water disposal	item	1.00	KSh 145,000	KSh 145,000
			Trade subtotal	KSh 625,000

Cost summary

Item	Amount
A — PRELIMINARIES	KSh 495,000
B — SUBSTRUCTURE	KSh 2,134,220
C — SUPERSTRUCTURE	KSh 9,209,052
D — ROOFING	KSh 1,632,950
E — FINISHES	KSh 1,573,056
F — JOINERY	KSh 523,500
G — ELECTRICAL	KSh 315,480
H — PLUMBING	KSh 625,000
Subtotal of works	KSh 16,508,258
Contingency 7.5 %	KSh 1,238,119
Professional fees 8 % (Architect + Engineer + QS)	KSh 1,320,661
Sub-total before VAT	KSh 19,067,038
VAT 16 %	KSh 3,050,726
GRAND TOTAL	KSh 22,117,764
Cost per m² (KES)	KSh 100,535 / m²

6. STATUTORY COMPLIANCE PACK

All approvals required under Kenyan law before construction. The contractor and professional team shall ensure each item is closed prior to mobilisation.

Approval	Authority / Statute	Documents required
1. County Planning Approval	County Govt (Building Plans Section)	Stamped architectural & structural drawings, location plan, ownership docs, NCA
2. NCA Project Registration	National Construction Authority Act 2011	NCA-1 form, contractor registration certificate, levy 0.5 % of contract sum.
3. NEMA Project Report (PR)	EMCA 1999, EIA Regs 2003	Required for projects > 5 ha or near watercourses; PR submitted to NEMA county
4. Public Health Approval	Public Health Act Cap. 242	Septic system & water supply review by County Public Health Office.
5. EPRA / KPLC Energy Connection	EPRA Regulations	Electrical drawings & wayleave application; final inspection by EPRA-registered C
6. Fire Safety Certificate	County Fire Marshal	Means-of-escape, extinguisher schedule, hydrant layout (NFPA 241 / Kenya Fire
7. DOSHS Notification	OSHA 2007 §41	Notification before commencement; site safety plan, induction registers, ISO 4500

6.1 Professional certification (signature pages)

Role	Name & Reg.	Signature & Stamp	Date
Architect (BORAQS)	— REQUIRED —	_____	Date: _____
Structural Engineer (EBK)	— REQUIRED —	_____	Date: _____
Quantity Surveyor (BORAQS)	— REQUIRED —	_____	Date: _____
Site Manager (NCA)	— to be appointed —	_____	Date: _____
Client	—	_____	Date: _____

I certify that the design has been carried out in accordance with the relevant Kenyan codes (NCA Act 2011, EMCA 1999, OSHA 2007) and applicable Eurocodes / British Standards, and that I take professional responsibility for the section of work above corresponding to my registration.

8. PROJECT SUMMARY

Phase phase_1: Site Analysis

water_table_depth	3.5
water_table_source	regional_estimate
water_table_note	Regional estimate from climate/soil model -- verify with on-site geotechnical su
soil_type	clay
vegetation_coverage	0.65
slope_angle	8.25
suitability_pct	90
nearby_structures	survey_required
land_use	residential
climate_zone	tropical
analysis_method	regional_data_model
data_source	EIMS Regional Database + GPS Coordinates

Phase phase_2: Geotechnical Design

bearing_capacity_kPa	150
soil_classification	CH
recommended_foundation	STRIP
settlement_mm	40
slope_stability	stable
water_table_depth	3.5
excavation_depth	2.8

Phase phase_3: Foundation Design

foundation_type	PAD
foundation_depth_m	1.3
footing_width_m	1.82
total_building_load_kN	1650.0
bearing_capacity_kPa	150
safety_factor	3.0
concrete_volume_m3	42.9
reinforcement_steel_kg	3861.0
soil_type	clay
water_table_factor	standard
design_standard	BS 8004 / Eurocode 7
bim_ready	True

Phase phase_4: Floor Plans

floor_count	2
total_area_sqm	220.0
bathrooms	3
rooms	{'bedrooms': 4, 'bathrooms': 3, 'living_areas': 1, 'kitchen': 1, 'extra_rooms':

style	modern
area_per_unit	220.0
building_code_compliance	True
area_efficiency_percent	80.0
cost_per_sqm_estimate	calculated_in_costing_phase

Phase phase_5: Electrical Design

total_connected_load_kW	9.38
max_demand_kW	5.63
diversity_factor	0.6
supply_type	single-phase 230V
main_current_A	24.5
lighting_points	12
socket_outlets	12
total_circuits	7
lighting_circuits	2
socket_circuits	2
fixed_appliance_circuits	3
distribution_boards	1

Phase phase_6: Plumbing Design

total_occupants	6
daily_demand_liters	900
peak_hourly_liters	90
total_fixture_units	34.5
main_supply_pipe_mm	32
cold_water_pipes	11
hot_water_pipes	7
waste_outlets	10
soil_stacks	1
hot_water_storage_liters	240
water_heater_kW	9.6
storage_tank_liters	1000

Phase phase_7: BOQ

material_items	77
total_material_cost	63292.76
currency	USD
items_by_category	{'mep_hvac': 5, 'structural': 17, 'external': 12, 'plumbing': 14, 'electrical':
items	[{'item_no': 1, 'description': 'Portland Cement (50kg)', 'unit': 'bag', 'quantit
calculation_basis	{'total_area_m2': 220.0, 'stories': 2, 'units': 1, 'bedrooms_per_unit': 4, 'foot
waste_factor_applied	True
method	quantity_takeoff_from_dimensions

Phase phase_8: Infrastructure

solar_system	{'panels_kw': 4.4, 'panels_count': 11, 'battery_kWh': 6.6, 'estimated_cost_usd':
borehole_system	{'depth_meters': 8, 'estimated_yield_lpm': 15, 'pump_kw': 3, 'estimated_cost_usd
grid_option	{'connection_cost_usd': 260, 'monthly_tariff_estimate_usd': 33}
recommendation	solar_grid_hybrid
estimated_annual_saving_usd	819.84

Phase phase_9: Landscape Design

lawn_area_m2	77.0
hardscape_area_m2	22.0
garden_area_m2	33.0
plants_count	264
irrigation_system	True
recommended_style	tropical_lush
estimated_cost_usd	759
maintenance_monthly_usd	15

Phase phase_10: Permits & Compliance

jurisdiction	Kenya
building_code	Kenya Building Code 2020
building_code_compliance	REQUIRES_VERIFICATION
permits_required	6
estimated_timeline_days	45
estimated_fees_usd	352
setback_compliance	{'front_m': 5, 'side_m': 2, 'rear_m': 3}
required_documents	['Architectural drawings (stamped)', 'Structural calculations (engineer-certifie

Phase phase_11: 3D Visualization

status	ready
model_endpoint	/api/drawings/3d-model

Phase phase_12: Live Costing

material_cost	8228058.8
labor_cost	925656.62
equipment_cost	658244.7
overhead	732297.23
contingency	1054425.74
subtotal	11598683.09
contractor_profit	1739802.46
total_project_cost	13338485.55
cost_per_sqm	60629.48
labor_multiplier	0.25
region	Kenya
currency	KES

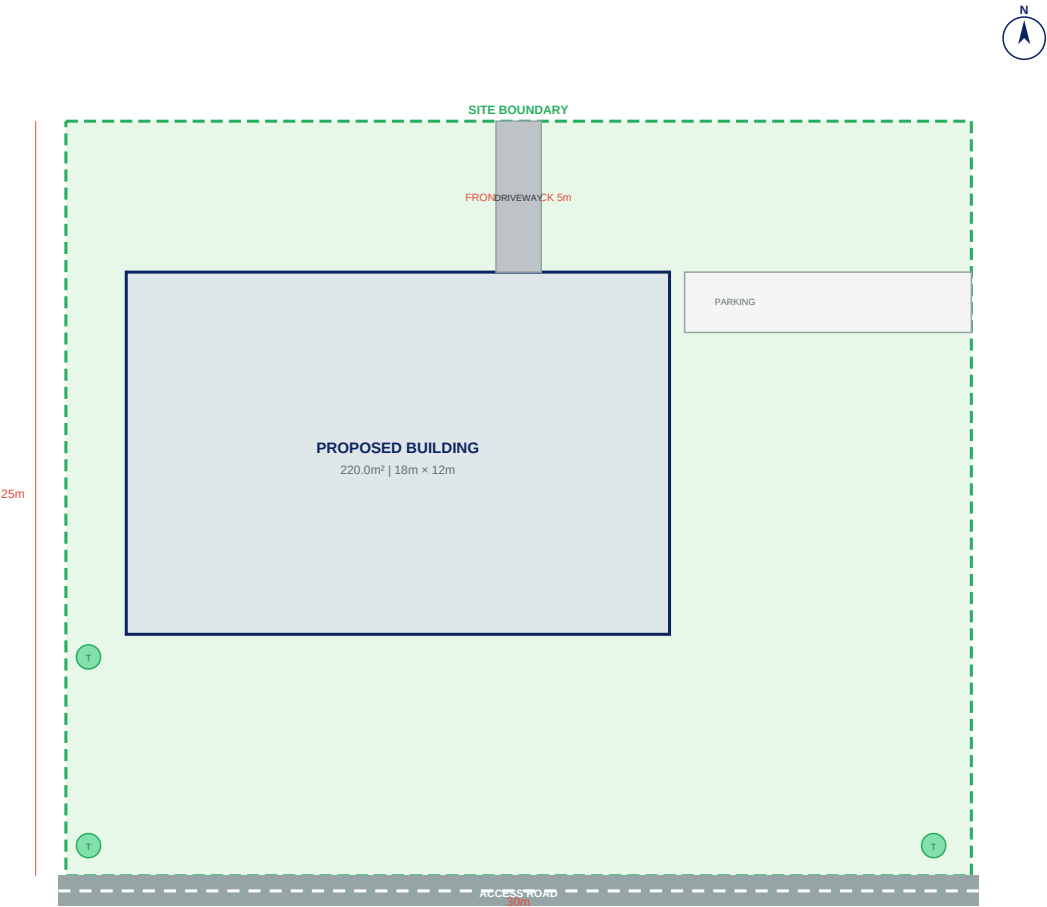
Phase phase_13: Integration Layer

status	complete
export_formats	['IFC2x3', 'FBX', 'DXF', 'PDF', 'XLSX']

9. SITE PLAN

Site layout showing building footprint, access roads, setbacks and landscaping. Area: 220 m²

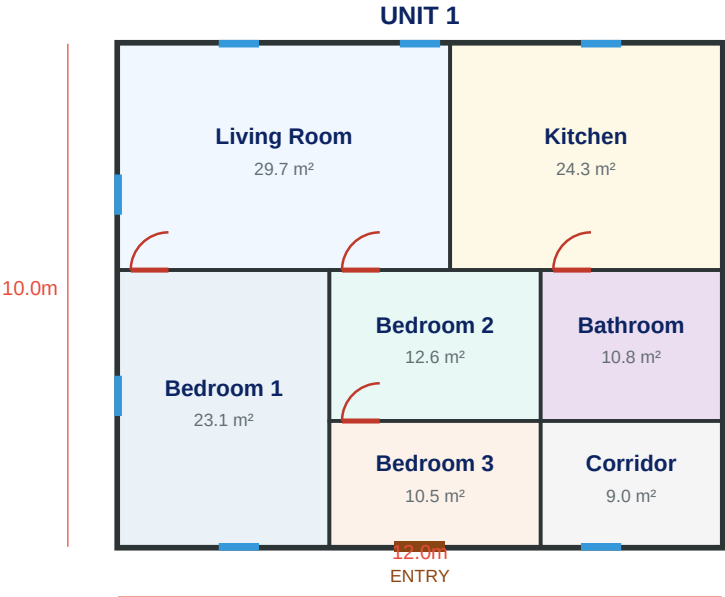
SITE PLAN - Nairobi, Kenya | Plot: 30m×25m



10. ARCHITECTURAL FLOOR PLAN

Ground floor layout — 1 unit(s), 4 bedrooms, showing room dimensions, doors, windows and circulation.

FLOOR PLAN - 220.0m² | 1 Units | 2 Storey(s)



EMERSON EIMS BUILDING SUITE PRO

Area: 220.0m² | 1 Units × 4 Bed

Floor Plan | Scale 1:50 | 2026-04-27 | Drawing: FP-001

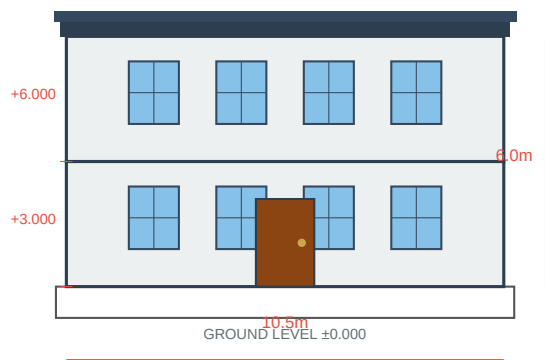
IFC Ready | BIM Level 2

11. BUILDING ELEVATIONS

4a. NORTH ELEVATION

Front view showing 2 storey(s), window/door openings, roof profile and floor levels.

NORTH ELEVATION - 2 Stories | Style: Modern



EMERSON EIMS | NORTH ELEVATION

Scale 1:50 | 2026-04-27 | Drawing: EL-N01

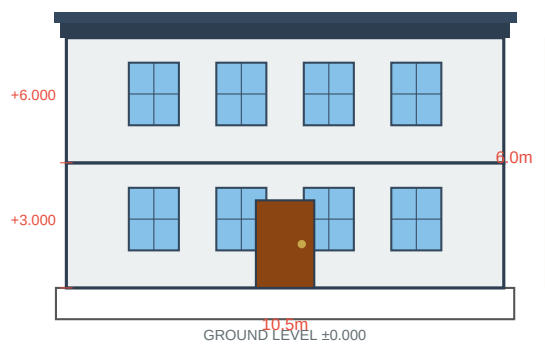
Stories: 2 | Height: 6.0m

Style: Modern | IFC Ready

4b. EAST ELEVATION

Side view showing building depth, roof slope and storey heights.

EAST ELEVATION - 2 Stories | Style: Modern



EMERSON EIMS | EAST ELEVATION

Scale 1:50 | 2026-04-27 | Drawing: EL-E01

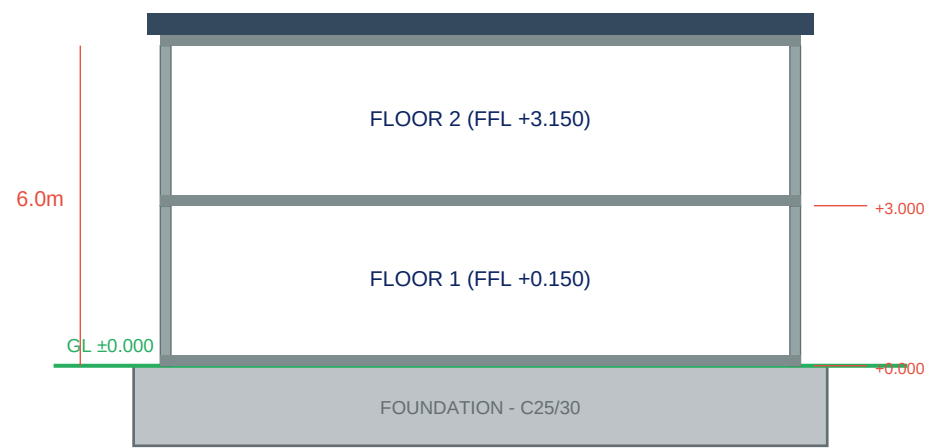
Stories: 2 | Height: 6.0m

Style: Modern | IFC Ready

12. BUILDING CROSS SECTION

Vertical cut through building showing floor slabs, storey heights (2 floors), foundation depth, roof structure and structural members.

CROSS SECTION A-A | 2 Stories



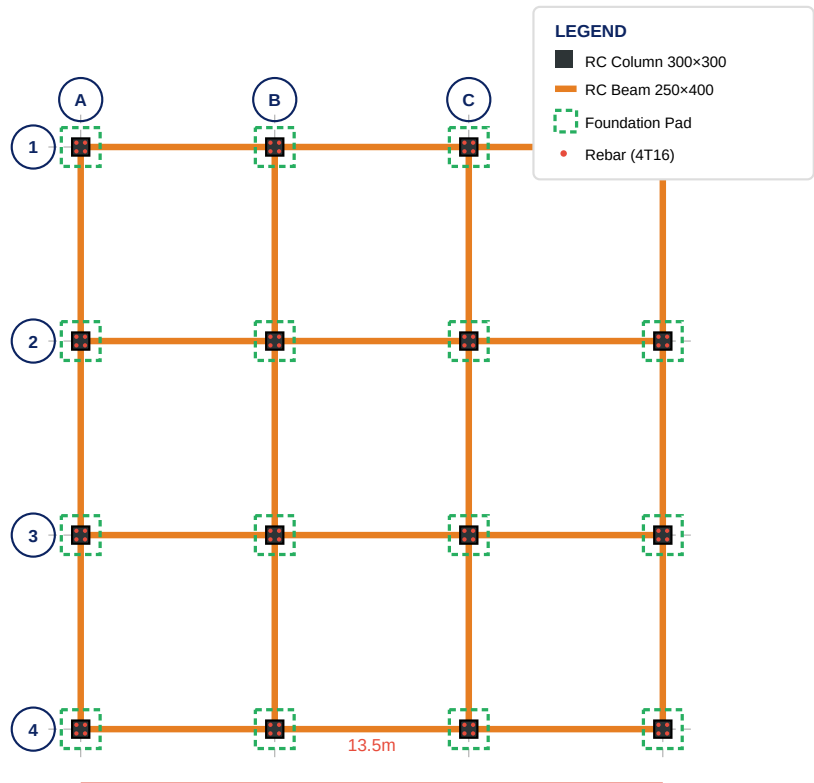
EMERSON EIMS | BUILDING SECTION A-A

2026-04-27 | SEC-001 | Scale 1:50

13. STRUCTURAL LAYOUT

Column grid, beam layout and foundation pads. Foundation type: Strip + Ground Beam. All dimensions in metres. Concrete grade C25/30, Steel Y16.

STRUCTURAL LAYOUT - Strip + Ground Beam Foundation | 2F | Grid 4.5m×4.5m



EMERSON EIMS | STRUCTURAL LAYOUT

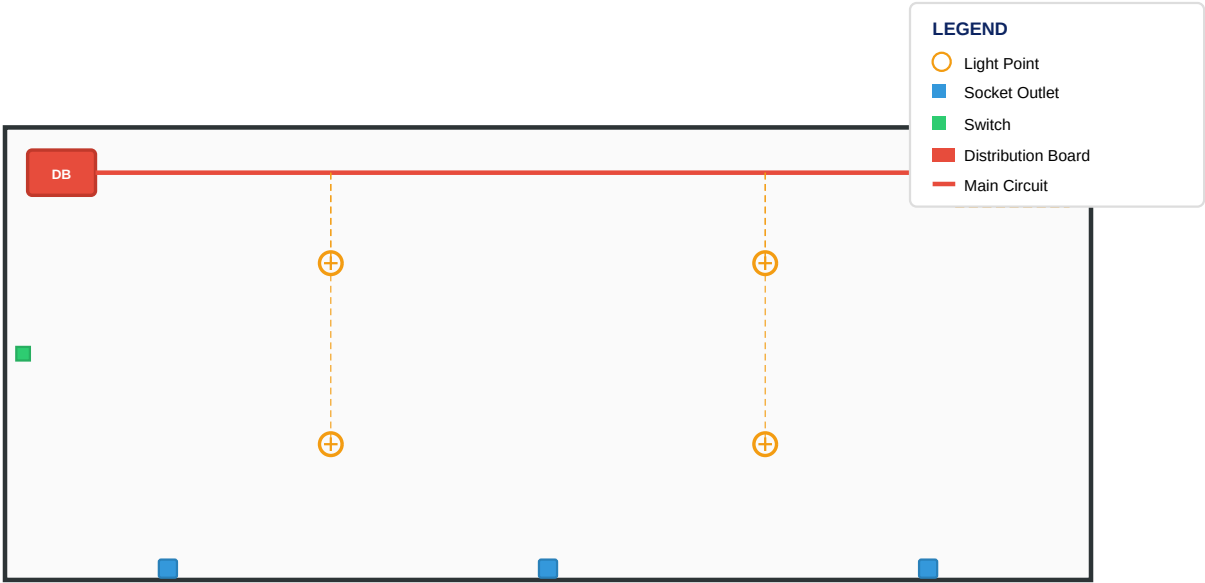
Grid: 4.5m × 4.5m | Columns: 16 | 2026-04-27 | STR-001

Concrete: C25/30 | Steel: Y16
Foundation: Strip + Ground Beam | FOS ≥ 3.0

14. MEP LAYOUT — ELECTRICAL

Electrical distribution layout showing DB positions, circuit runs, socket/light points and earthing.

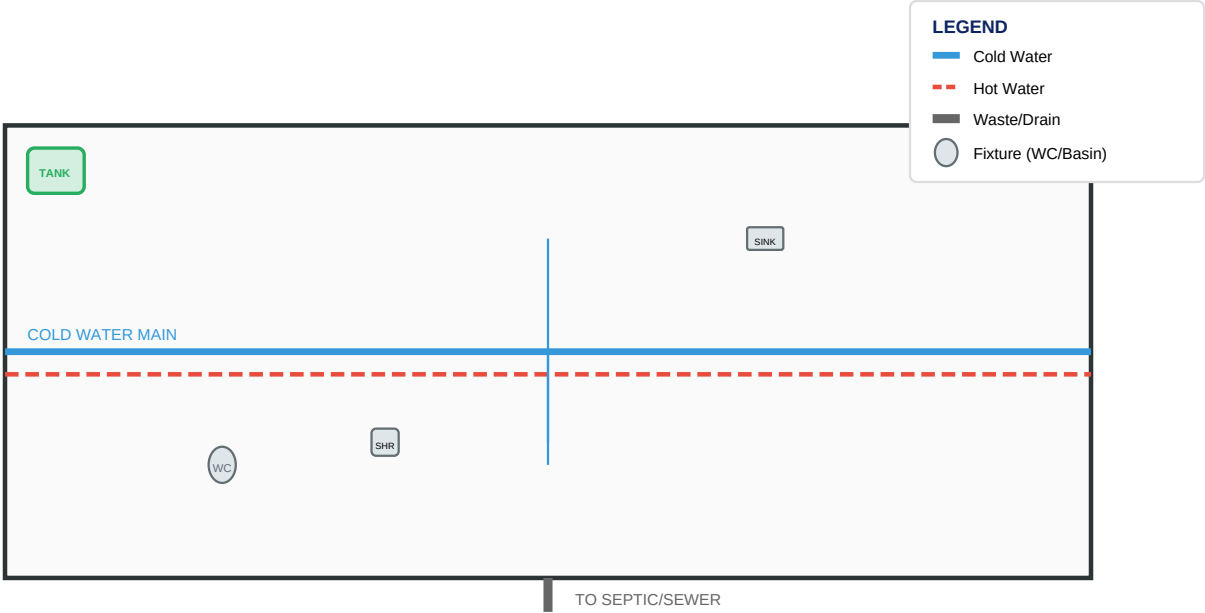
ELECTRICAL LAYOUT - 220.0m² | 1 Units



15. MEP LAYOUT — PLUMBING

Plumbing layout showing supply/drain runs, fixture locations, water heater and main shutoff valve.

PLUMBING LAYOUT - 220.0m² | 1 Units



16. COST SUMMARY

Item	Amount (KES)
Material Cost	KSh 8,228,058.80
Labor Cost	KSh 925,656.62
Equipment Cost	KSh 658,244.70
Overhead	KSh 732,297.23
Contingency	KSh 1,054,425.74
Subtotal	KSh 11,598,683.09
Contractor Profit	KSh 1,739,802.46
Total Project Cost	KSh 13,338,485.55
Cost Per Sqm	KSh 60,629.48
Labor Multiplier	KSh 0.25
Fx From Usd	KSh 130.00

This report was generated by EMERSON EIMS Building Suite Pro. All drawings are indicative and must be verified by a licensed engineer before construction.